PRACTICAL – 1

AIM: Perform a practical to study different hypervisors for windows and Linux machine. Perform a practical to create virtual machine using type-2 hypervisor in windows.

- A hypervisor is a form of virtualization software used in Cloud hosting to divide and allocate the
- resources on various pieces of hardware. The program which provides partitioning, isolation, or abstraction is called a virtualization hypervisor. The hypervisor is a hardware virtualization technique that allows multiple guest operating systems (OS) to run on a single host system at the same time. A hypervisor is sometimes also called a virtual machine manager(VMM).

Types of Hypervisors:

Type-1 Hypervisor (Bare-Metal)

- This type runs directly on the physical hardware, without a host operating system. It manages the hardware and allows multiple operating systems (VMs) to run on top of it.
- **Use case**: Preferred in enterprise environments and data centers because it offers better performance, scalability, and direct control over hardware.
- **Examples**:
  VMware ESXi

  Microsoft Hyper-V

  Xen

  KVM (Kernel-based Virtual Machine)
- **Advantages**:
  High performance and efficiency.

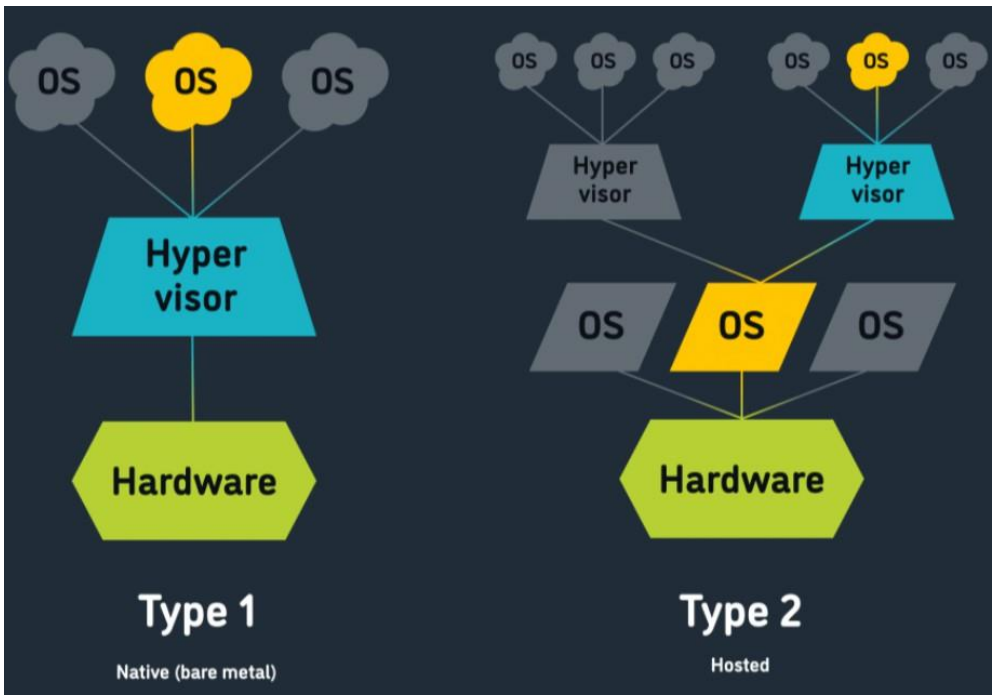
  Direct access to hardware, making it faster.

  Used in large-scale server environments.
- **Disadvantages**:
  More complex to set up and manage.

  Requires dedicated hardware.

Type-2 Hypervisor (Hosted)

- This type runs on top of an existing host operating system (e.g., Windows, Linux, macOS). It relies on the host OS to manage hardware resources while managing the VMs.
- **Use case**: Commonly used on desktop systems for testing, development, or running multiple OS environments.
- **Examples**:
  - Oracle VirtualBox
  - VMware Workstation
  - Parallels Desktop (for macOS)
- **Advantages**:
  - Easier to set up and use.
  - Good for personal or small-scale environments where direct access to hardware isn't crucial.
- **Disadvantages**:
  - Slower performance due to dependence on the host OS.
  - Not suitable for large-scale environments or high-performance needs.



- **Type-1 Hypervisor**: Hardware → Hypervisor → Virtual Machines
- **Type-2 Hypervisor**: Hardware → Host OS → Hypervisor → Virtual Machines

- Creating a Virtual Machine with VirtualBox:

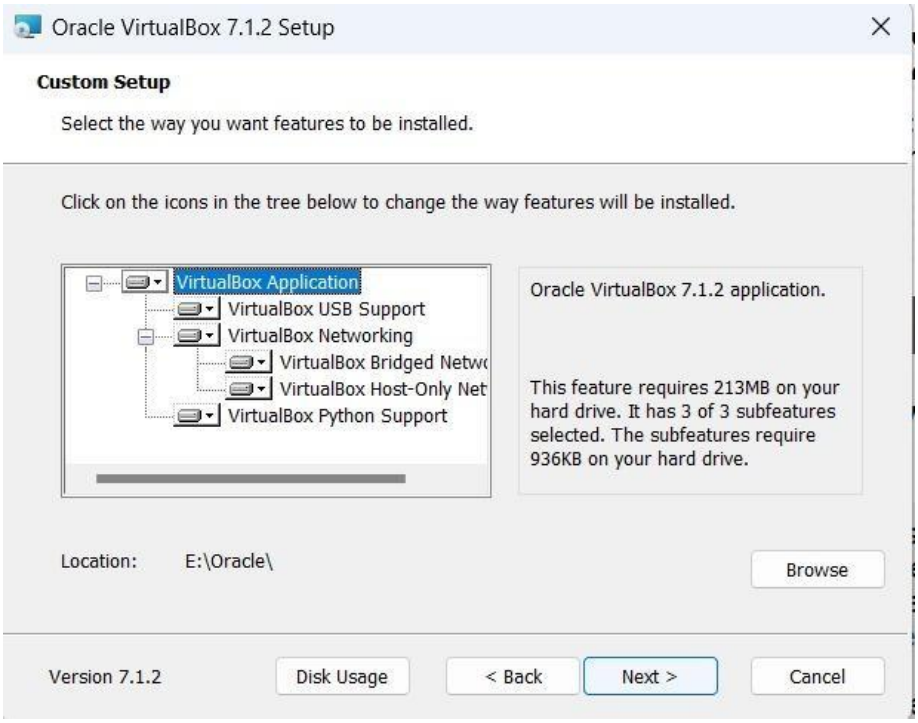
Download and Install VirtualBox:

1. **Download VirtualBox:** Visit the VirtualBox website and navigate to the downloads section to download VirtualBox.
2. **Install VirtualBox:** Locate the downloaded file and follow the installation instructions to install VirtualBox on your system.

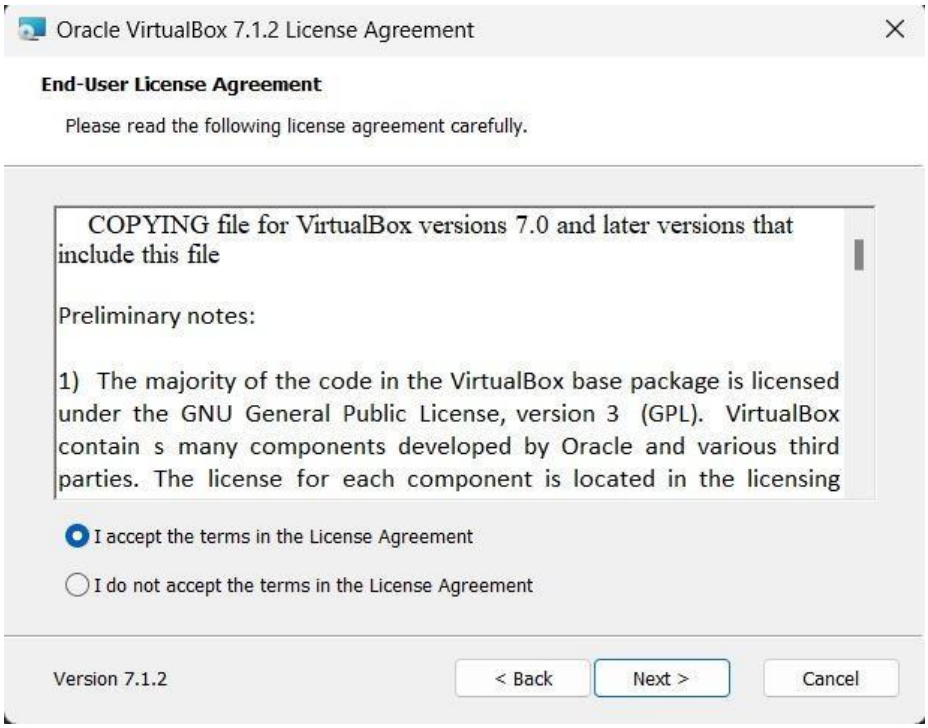
Step 1: Begin by running VirtualBox on your system. You'll see the Oracle VM VirtualBox 7.0.18 Setup Wizard. Follow the on-screen instructions to proceed with the installation.



Step 2: Click the "Next" button to proceed. In this step, you will need to select the installation location for VirtualBox. Choose a directory where you want the software to be installed and then continue to the next step.



Step 3: Click the "Next" button to proceed through the setup wizard. Continue following the prompts until you reach the installation options. Make sure to review each step carefully. Complete the setup by selecting the appropriate installation settings.

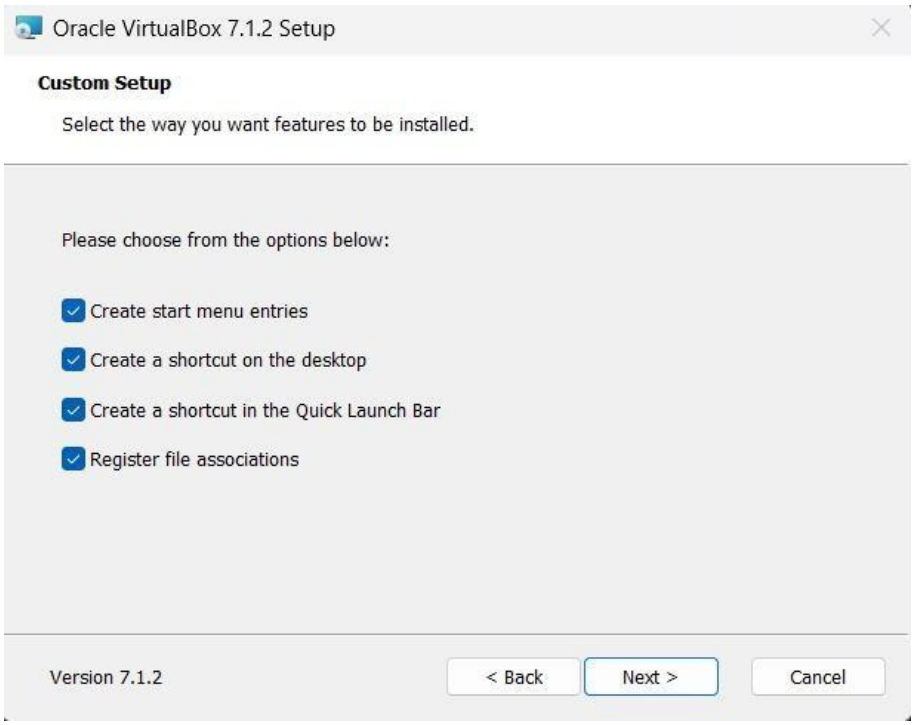


Step 4: Click the "Yes" button when prompted. You will see a warning about network interface changes. This warning is to inform you that VirtualBox will modify your network settings.

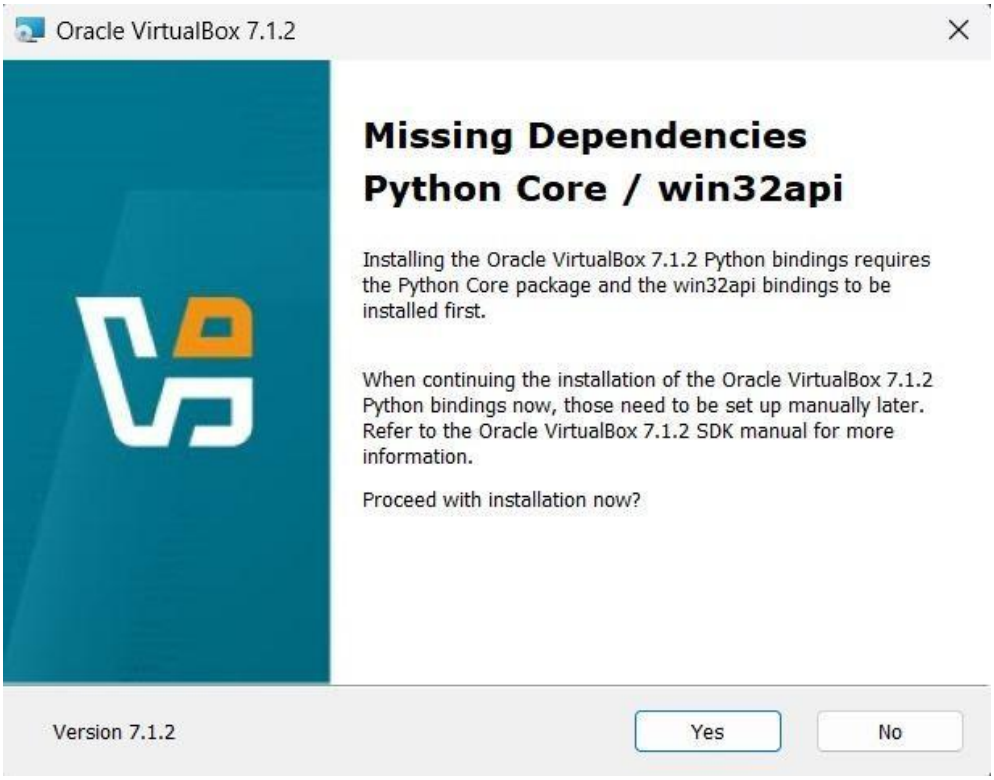
Proceed by confirming the changes to continue with the installation.



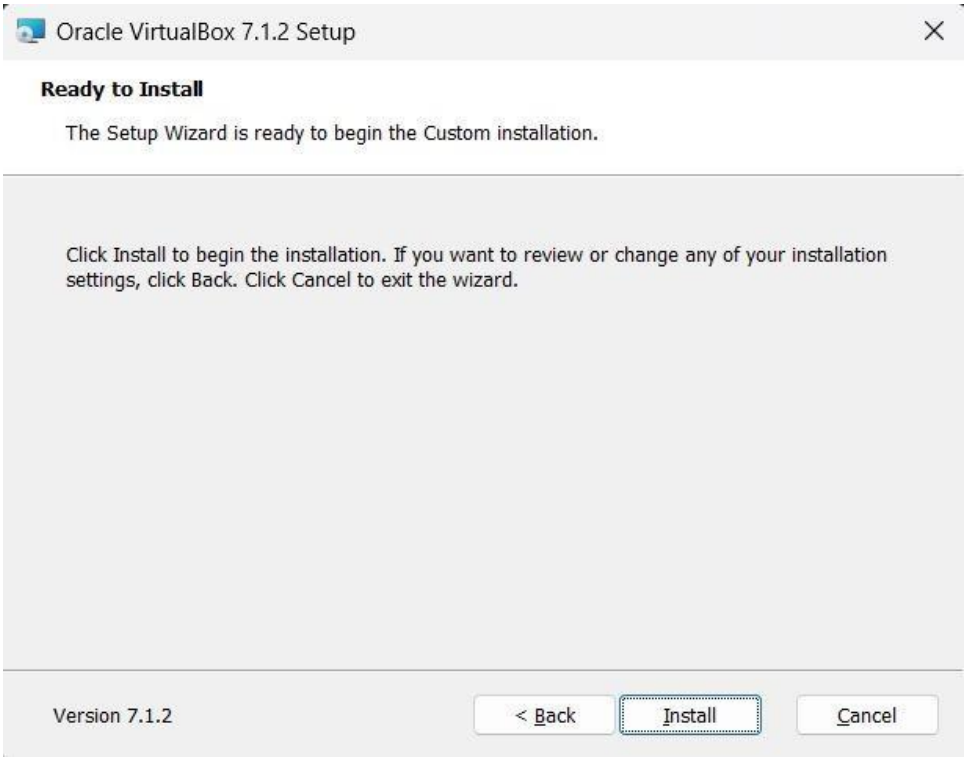
Step 5: Click the "Next" button to proceed through the setup wizard. Continue following the prompts until you reach the installation options. Make sure to review each step carefully. Complete the setup by selecting the appropriate installation settings.



Step 6: Click the "Yes" button to proceed through the setup wizard.



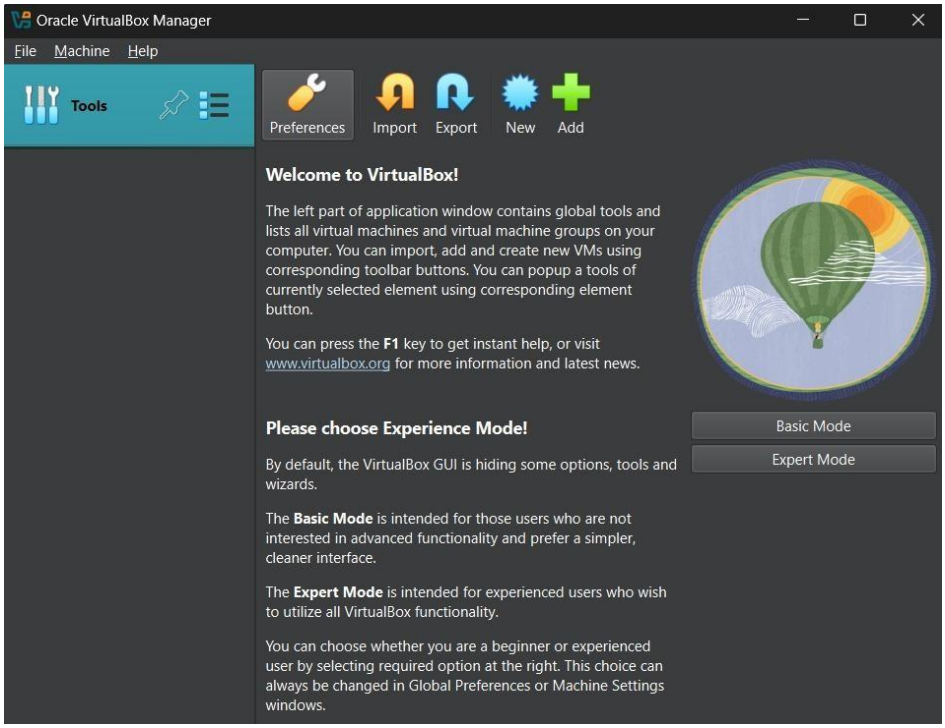
Step 7: Click the "Install" button to start the installation process. The setup will now copy files and configure VirtualBox on your system. Wait for the installation to complete, which may take a few minutes.



Step 8: Click the "Finish" button to exit the Setup Wizard. This will close the installation window and complete the setup process for VirtualBox. You can now start using the software.



Step 9: Open VirtualBox by finding its icon on your desktop or in the applications menu of your system. Double-click the icon to launch the application. The VirtualBox main window will appear, where you can start creating new virtual machines or manage existing ones. Explore the options and settings to configure your virtual environment according to your needs.



Download Ubuntu and open in Virtual Box :

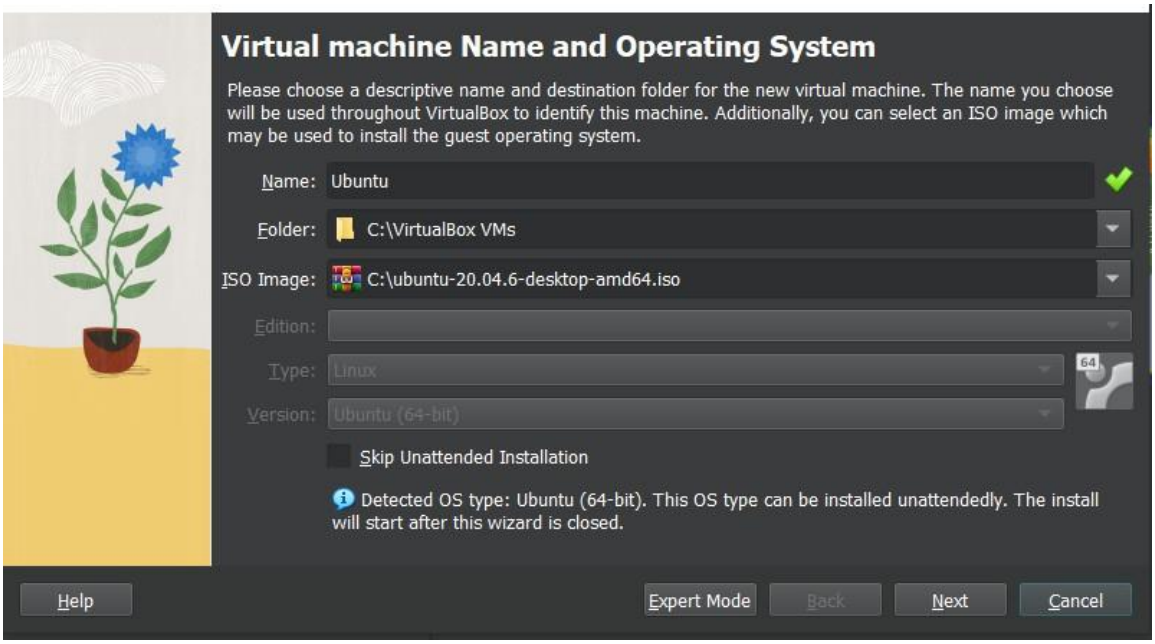
- **Go to this page and download Ubuntu iso file :**
  https://ubuntu.com/download/desktop

  On to this web page download Ubuntu iso file.

Step 9 - Follow the below Step to start Ubuntu in Virtual box:

click on the "New" button located in the toolbar. This will start the process of creating a new virtual machine. Follow the prompts to configure your new virtual machine according to your requirements.

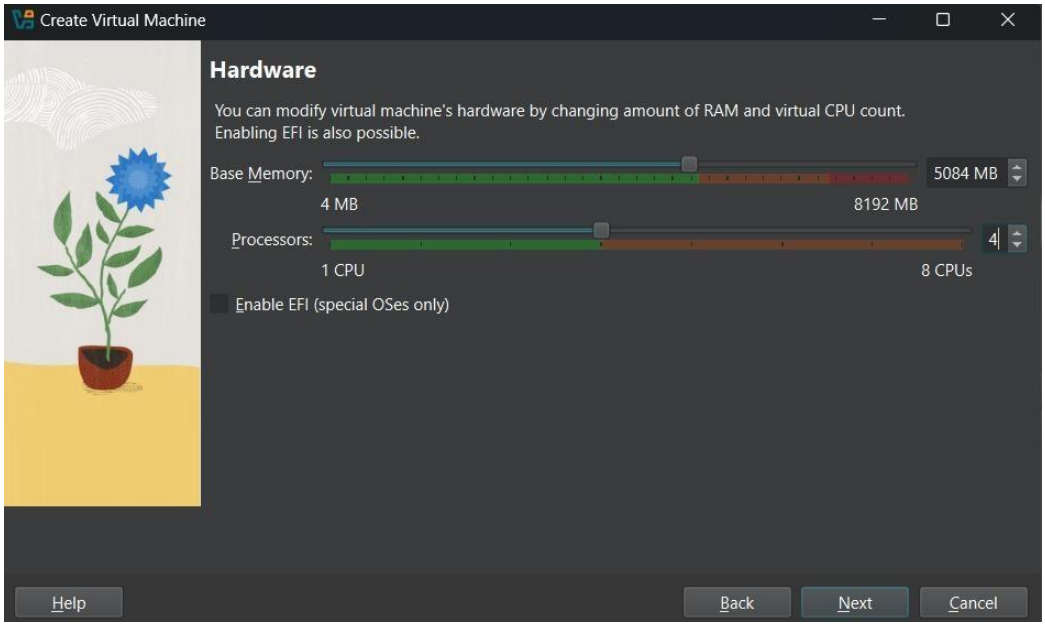
Step 10: When you click "New," a setup box will open. In this box, you will need to enter a name for your virtual machine. Choose a descriptive name for your host. Next, specify the folder where you want to store the virtual machine files. You will also need to select the ISO image for installation. For this setup, choose the edition, type, and version of Ubuntu Linux that you are using. Follow the prompts to complete the configuration. Upload the Ubuntu ISO file that you have previously downloaded. In the setup box, look for the option to browse and select the ISO file. Choose the file from your local storage and upload it. Once the ISO file is uploaded, click the "Next" button to proceed with the setup. This will allow you to configure additional settings for the virtual machine.



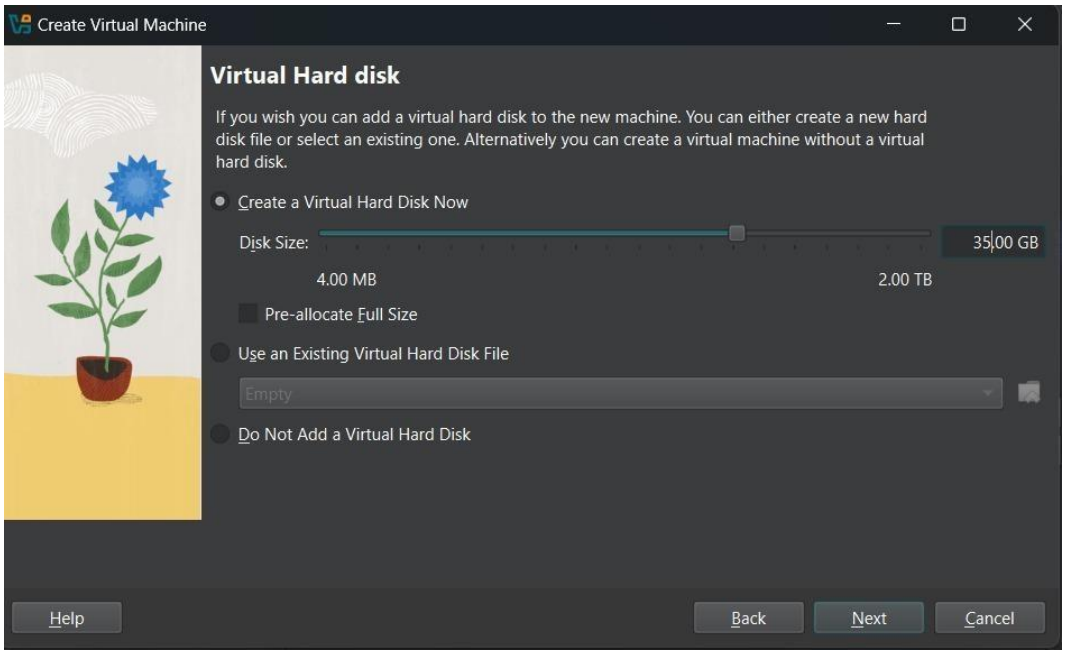
Step 11: A new box will appear prompting you to enter your user credentials. Input the desired username and password for the virtual machine. This information will be used to log in to the Ubuntu system once the installation is complete. After entering the credentials, click the "Next" button to continue with the setup process.



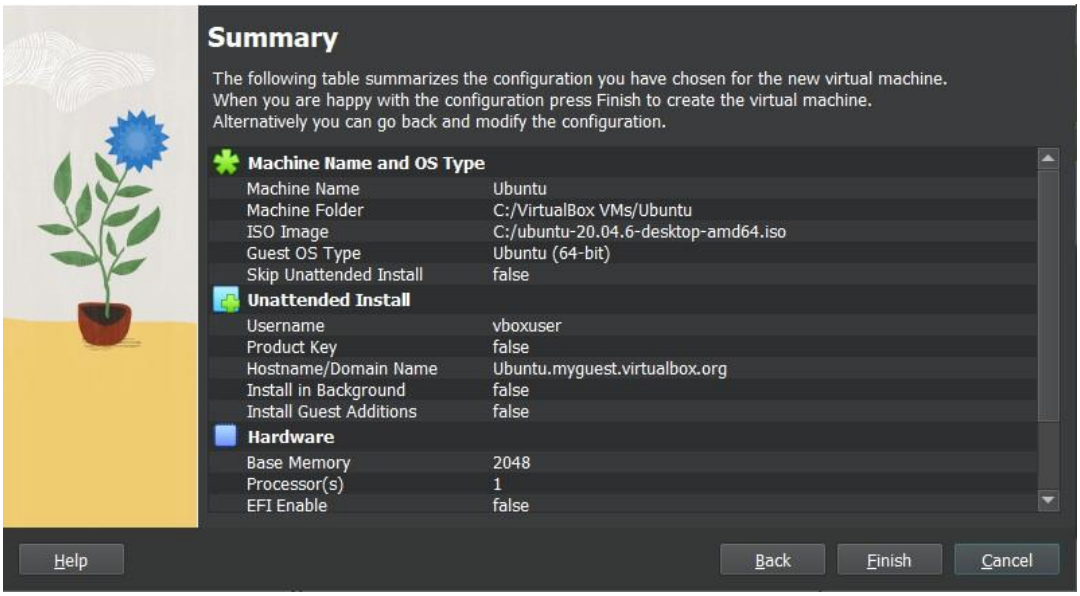
Step 12: A new box will appear asking you to set the size of the Base Memory (RAM) for your virtual machine. Adjust the slider or enter the amount of RAM you want to allocate based on your requirements and the resources available on your host system. After setting the desired amount of RAM, click the "Next" button to proceed with the configuration.



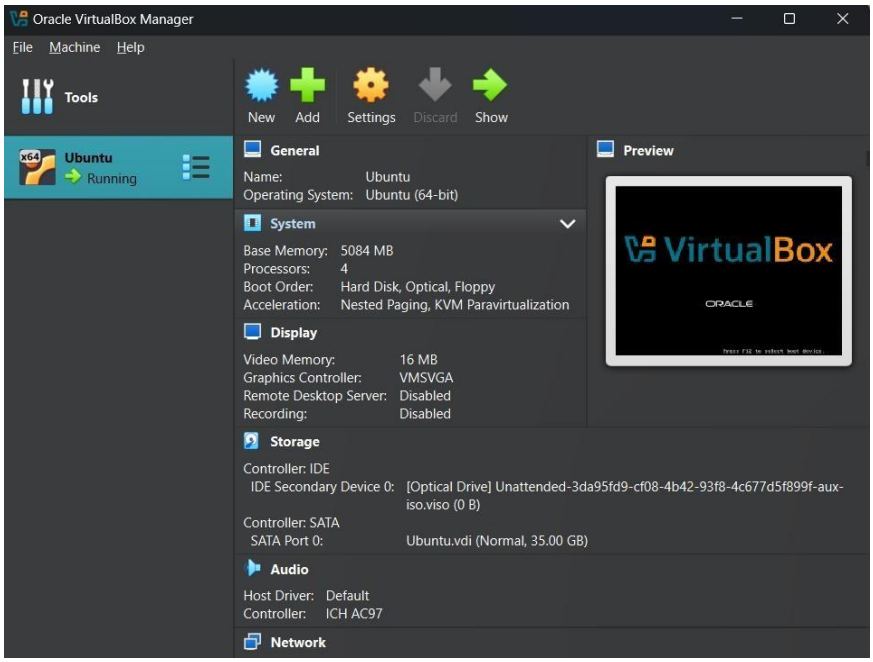
Step 13: A new box will appear where you need to set the Disk Size (ROM) for your virtual machine. Adjust the size of the virtual hard disk according to your storage needs. You can select the amount of disk space to allocate for the virtual machine. Once you have set the desired disk size, click the "Next" button to continue with the setup process.



Step 14: A new box will open displaying a summary of your virtual machine configuration, including the settings for memory, disk size, and other parameters. Review the summary to ensure all settings are correct. If everything looks good, click the "Finish" button to complete the creation of your virtual machine. This will finalize the setup and return you to the main VirtualBox window.



Step 15: In the VirtualBox main window, select your newly created virtual machine from the list. Click on the "Start" button to begin the virtual machine. The Ubuntu installation process will start, and the virtual machine will boot up. Follow any additional prompts to complete the installation and setup of Ubuntu within the virtual environment.



Step 16: Wait for a few moments while Ubuntu boots up within the virtual machine. This process may take some time as the operating system loads and initializes. Once the boot process is complete, you will see the Ubuntu login screen or desktop, depending on the stage of the installation or setup process.

